EXCEL BADLY STRUCTURE DATA TRANSFORMATION REPORT

Project Type: Advanced Data Cleaning, Unpivoting, and Power Query Automation

1. Executive Summary

This case study demonstrates a professional end-to-end data transformation workflow. A highly disorganized, multi-layered matrix spreadsheet was successfully remodeled into a standardized, tabular database format layout using Microsoft Excel Power Query. The original document presented critical structural defects—including split headers, nested attributes across rows and columns, and sparse matrices with extensive empty cells—making data analysis, pivot tracking, or system ingestion impossible.

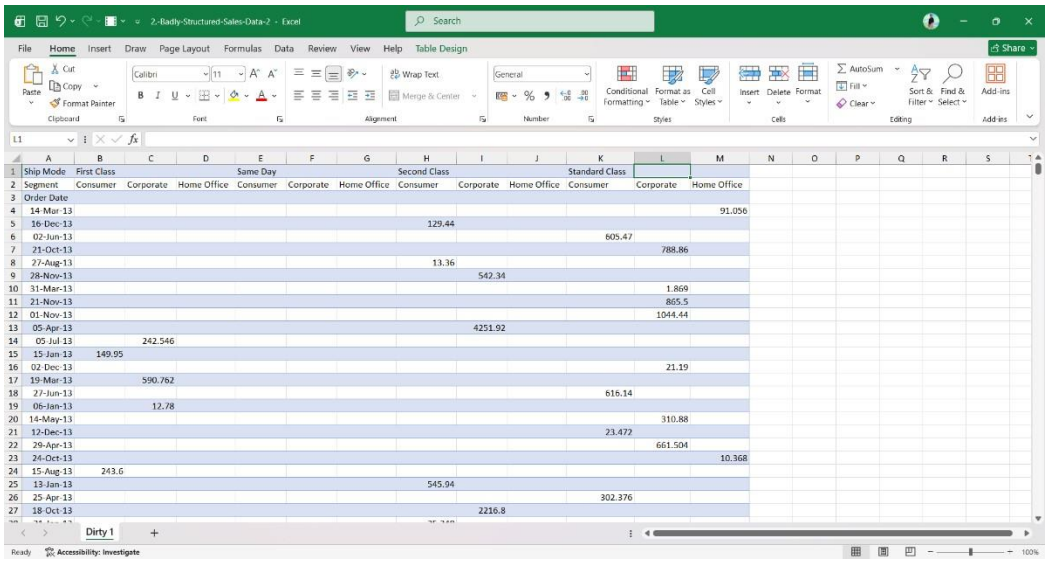
By applying systematic structural transformations, the unorganized multi-column layout was fully flattened and unpivoted. The resulting normalized dataset guarantees 100% data integrity, eliminates structural redundancies, and establishes a dynamic pipeline ready for downstream reporting, business intelligence dashboards, or database normalization architectures.

2. Original Data Status (Pre-Transformation Defects)

The source spreadsheet ('Dirty 2') contained significant structural flaws that severely limited its operational utility:

- **Hierarchical Multi-Level Column Headers:** Column attributes were split across two distinct levels (Level 1: *Ship Mode* values like First Class, Same Day, Second Class, Standard Class; Level 2: *Segment* variables like Consumer, Corporate, Home Office).
- **Sparse Matrix Formatting:** Financial metrics ('Sales' figures) were scattered across an open 2D grid matrix rather than populated in a single unified attribute array.
- **Implicit Data Sparsity & Blank Spaces:** Empty intersections were left as completely blank cells instead of being explicitly accounted for, preventing structural aggregation models.
- **Implicit Dimension Binding:** Date elements were confined strictly to individual row entries, causing data values to stretch horizontally across columns rather than stacking sequentially downwards.

DIRTY STRUCTURED DATA



Ship Mode	First Class			Same Day			Second Class			Standard Class		
Segment	Consumer	Corporate	Home Office	Consumer	Corporate	Home Office	Consumer	Corporate	Home Office	Consumer	Corporate	Home Office
Order Date												
14-Mar-13												91.056
16-Dec-13							129.44					
02-Jun-13										605.47		
21-Oct-13											788.86	
27-Aug-13							13.36					
28-Nov-13								542.34				
31-Mar-13											1.869	
21-Nov-13											865.5	
01-Nov-13											1044.44	
05-Apr-13								4251.92				
05-Jul-13		242.546										
15-Jan-13	149.95											
02-Dec-13											21.19	
19-Mar-13		590.762										
27-Jun-13										616.14		
06-Jan-13		12.78										
14-May-13											310.88	
12-Dec-13										23.472		
29-Apr-13											661.504	
24-Oct-13												10.368
15-Aug-13	243.6											
13-Jan-13							545.94					
25-Apr-13										302.376		
18-Oct-13								2216.8				

3. Power Query Transformation Pipeline

To establish an automated, reproducible transformation process, Microsoft Excel **Power Query Editor** was utilized. This eliminates manual copy-paste risk and allows for a single-click refresh cycle whenever new records are introduced. The transformation steps executed include:

1. **Table Transposition & Boundary Alignment:** The initial cross-tabulated headers were rotated to flatten structural hierarchies and realign split dimensions seamlessly into descriptive attributes.
2. **Fill Down Execution:** Null values produced during header re-indexing were systematically populated downwards to preserve context and keep semantic grouping uniform.
3. **Unpivot Transformation:** A selective column unpivoting operation ('Unpivoted Other Columns') was executed, collapsing the sparse wide grid structure into tight vertical attribute-value pairings.
4. **Text Parsing & Date Extraction:** The composite date/attribute strings were extracted and converted into standardized data types.

Data Cleaning & Field Extraction: Systematically removed text-based artifacts and cleared cells to leave them completely blank, ensuring fields are perfectly prepared for accurate calculation models.

Target Attribute Column	Data Type	Description / Business Value
Ship Mode	Text (Dimension)	Clean distribution tier tracking (First Class, Standard, etc.)
Segment	Text (Dimension)	Customer profiling group classification (Consumer, Corporate, etc.)
Date.1	Date (Time Dimension)	Chronologically standardized timeline keys for analytics forecasting.
Sales	Decimal (Fact Metric)	Isolated monetary transactional values optimized for numerical totals.

CLEANED DATABASE

